

Multi-Subject Engineering Practice Paper:

Theory/Long Answers

Q1. (Electrical) State and explain Kirchhoff's Current Law (KCL) and Voltage Law (KVL) with a circuit diagram.

Q2. (CAD) Differentiate between Wireframe, Surface, and Solid modeling in CAD. Which is best for mass property calculations?

Q3. (Math) Solve the differential equation $dy/dx + y/x = x^2$ using the Integrating Factor method.

Q4. (Mechanics) Explain the Work-Energy theorem. How does it apply to a block moving on a surface with friction?

Q5. (Physics) Derive the expression for De Broglie wavelength. Explain its significance in quantum mechanics.

Q6. (Materials) Draw and explain the typical stress-strain curve for a ductile metal. Label the yield point and ultimate tensile strength.